

Project in Data Science

Skin Lesion Classification

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1 Introduction

Skin cancer is a major public health concern, and early detection is essential for improving treatment outcomes. The most common forms include melanoma (MEL), basal cell carcinoma (BCC), and squamous cell carcinoma (SCC) [5]. Since access to dermatologists is limited in many regions, automated decision-support tools may help identify suspicious lesions earlier and support clinical screening.

This project investigates whether machine learning can distinguish between cancerous and non-cancerous skin lesions using interpretable image features. The feature extraction process is based on the ABC criteria commonly used in dermatology: asymmetry, border irregularity, and colour variation. These features were chosen because they are clinically meaningful and can be extracted from segmented lesion images.

The project was developed iteratively. First, we explored the dataset and checked the availability of segmentation masks. Then, we extracted ABC-based features and used them to train classification models. Finally, we investigated whether additional preprocessing and feature engineering could improve performance, especially in relation to image artifacts such as hair.

2 Data

The project uses the PAD-UFES-20 dataset [3], which contains clinical images of skin lesions together with patient metadata. The dataset includes six diagnostic categories: Actinic Keratosis (ACK), Basal Cell Carcinoma (BCC), Malignant Melanoma (MEL), Melanocytic Nevus (NEV), Squamous Cell Carcinoma (SCC), and Seborrheic Keratosis (SEK).

For this project, the task was simplified into binary classification. The diagnostic labels were grouped into cancerous and non-cancerous classes:

- **Cancerous:** BCC, SCC, and MEL.
- **Non-cancerous:** ACK, NEV, and SEK.

This grouping allowed the models to focus on separating potentially malignant lesions from the remaining cases. Each image was assigned a binary target value, where cancerous lesions were labelled as 1 and non-cancerous lesions as 0.

2.1 Methodology

The methodology followed a structured pipeline from data preparation to model evaluation. The main steps were:

1. Inspect the metadata and diagnostic labels.
2. Check whether each image had a corresponding segmentation mask.
3. Extract lesion-based features from the masked image

region.

4. Train classification models using the extracted features.
5. Evaluate model performance using accuracy, precision, recall, F1-score, and AUC.

The segmentation masks are important because the features should describe the lesion itself rather than the surrounding skin or background. Asymmetry and border features depend directly on the mask shape, while colour features are calculated only from pixels inside the lesion area.

3 Data Exploration

Before training any models, we conducted a systematic exploration of the PAD-UFES-20 dataset to assess its suitability for the classification task. This involved three aspects: the distribution of diagnostic labels and their binary groupings, the availability of segmentation masks required for ABC feature extraction, and the presence of missing values in the metadata. The dataset comprised 2103 skin lesion images, each associated with a diagnostic label and patient-related metadata. Understanding the structure of this data early in the pipeline was essential, as imbalances in class distribution, gaps in mask coverage, or missing feature values could each compromise the reliability of the trained models.

3.1 Diagnostic Class Distribution

Although the binary classes were almost balanced, the original diagnostic categories were unevenly distributed. BCC and ACK were the most common diagnoses, while MEL had the fewest samples. This is important because the binary task may appear balanced even though some clinically important categories, especially melanoma, are underrepresented.

Table 1: Distribution of diagnostic classes in the dataset.

Diagnosis	Number of images
BCC	815
ACK	634
NEV	220
SEK	201
SCC	184
MEL	49

After grouping the diagnoses, the dataset contained 1048 cancerous images and 1055 non-cancerous images. This near-even split reduces the risk of the model simply learning to predict the majority class.

Table 2: Distribution of binary target classes.

Class	Number of images
Non-cancerous	1055
Cancerous	1048

3.2 Mask Availability

The feature extraction process relies on segmentation masks to isolate the lesion from the rest of the image. Therefore, we checked whether every image had a matching mask file. The expected mask filename was created by adding `_mask` before the file extension. For example, the image

`PAT_1516_1765_530.png`

would be matched with

`PAT_1516_1765_530_mask.png`.

Images without masks cannot be processed in the same way as images with masks. Since the ABC features depend on the lesion region, missing masks were excluded.

3.3 Missing Values

Missing values were identified in several columns of the metadata. Since incomplete data can affect feature extraction, model training, and the fairness of model comparisons, rows with missing values in important feature columns were removed before training. In addition, images without corresponding segmentation masks were excluded, as features based on lesion shape, border, and colour could not be extracted reliably from them. Masks that were completely black were also removed, because they did not contain any visible lesion area and therefore could not provide meaningful segmentation information. This ensured that all models were trained and evaluated using the same complete and usable dataset.

4 Baseline classification Model

The extracted features were designed to reflect visual patterns associated with the ABC criteria.

4.1 Feature extraction

We extracted a set of interpretable descriptors from each lesion image and its corresponding segmentation mask. We designed these features to capture the visual properties associated with the ABC criteria used in clinical dermatology: asymmetry, border irregularity, and colour variation. We computed each feature programmatically and stored it as a numerical value to be used as input to our classification models.

4.1.1 Asymmetry

We calculated asymmetry from the binary lesion mask. The mask was split along horizontal and vertical axes, and the two halves were compared after mirroring. A larger mismatch indicates a more asymmetric lesion which can associate with dangerous skin lesions.

When faced with a binary mask such as `M`, we first defined the midpoints, split `M` into halves, flip one half on the another

and compute:

$$A = \frac{\sum XOR_H + \sum XOR_V}{2 \cdot \sum M}$$

where $\sum M$ is the total number of processed pixels and XOR_H and XOR_V are the horizontal and vertical XOR results that we get. Our score must be between $[0,1]$, when we get closer to 0, this means we are getting closer to perfect symmetry.

After our inspection, we realised that even though asymmetry is a key factor on detecting the skin lesions, lesions with high asymmetry score can be non-cancerous.

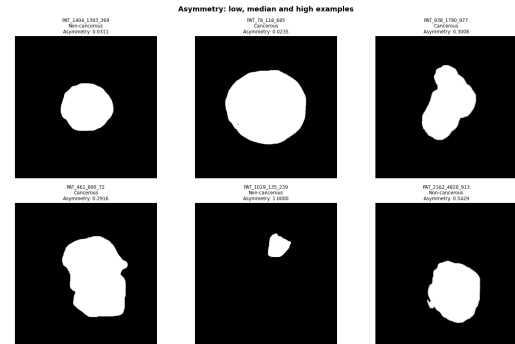


Figure 1: Asymmetry scores

4.1.2 Border

For the border features we used two key measures which are Convexity and Compactness.

$$\text{Compactness} = \frac{4\pi \cdot \text{Area}}{\text{Perimeter}^2}$$

Having a value close to 1 indicates that we have a circular shape. While lower value means that we have more irregular borders, which are more common in harmful skin lesions. The area is the scaled pixel count and perimeter is estimated by subtracting the mask from the original picture.

The second feature we used for the border was convexity, Which is the ratio of the lesion area over the convex hull area.

$$\text{Convexity} = \frac{\text{Area}}{\text{Area}_{\text{hull}}}$$

Values that are closer to one demonstrates a perfectly convex lesion, meanwhile the lower values reflect concavities and tracing at the borders, which might be a sign for a cancerous lesion.

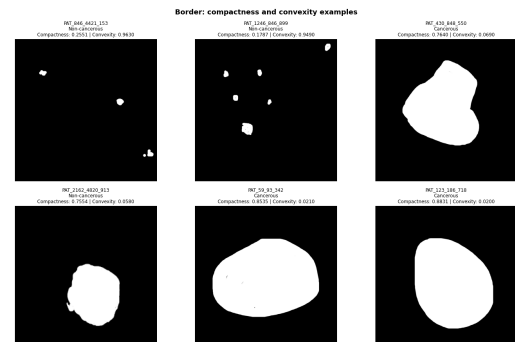


Figure 2: Border scores

4.1.3 Border and Asymmetry Analysis

After extracting both border and asymmetry features, we analyzed their distributions across cancerous and non-cancerous lesions to evaluate their separability. Plotting the two features against each other revealed a great overlap between the two classes, suggesting that these shape-based features capture clinically relevant information but are most effective when combined with additional descriptors rather than used in isolation.

This is consistent with how the ABC criteria are applied in clinical practice. So dermatologists do not rely on a single indicator, but instead consider asymmetry, border, and colour together when assessing a lesion. A lesion may score high on asymmetry or border irregularity for reasons unrelated to malignancy, such as anatomical variation or image artefacts from imprecise segmentation masks.

We therefore treat asymmetry and border as important components of a broader feature set.

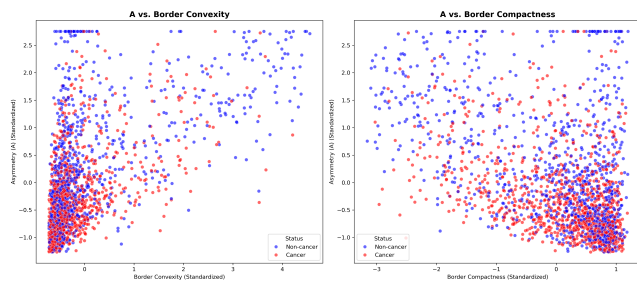


Figure 3: Asymmetry vs Border

4.1.4 Colour Analysis and Variation

Using the SLIC (Simple Linear Iterative Clustering) [1] Segmentation methods, our algorithm groups an image's pixels into uniformly sized clusters called superpixels. This helps to make pixels more meaningful by simplifying complex images while adhering to natural boundaries. The SLIC method places a grid of "centers" across the image, then iteratively shifts these centers to group similar adjacent pixels. After segmenting the image with SLIC, we extract the Hue, Saturation, and Value variances of the pixels inside the lesion mask of the images. We chose HSV specifically, rather than RGB, as it is more closely aligned with the way human visual system perceives color-making attributes [6]. In an HSV image, the colors are represented by Hue and Saturation, whereas the brightness of the pixels is reflected by Value [4]. HSV explicitly separates chromatic information (hue and saturation) from luminance (value), whereas in RGB, brightness is implicitly distributed across all three channels rather than represented independently. This allows imaging software to maintain color consistency across each lesion, even when dealing with uneven skin curvature, shadows, or changing lighting conditions. Looking closer at the statistical variances in the Hue of the skin pixels, our algorithm can quickly define the edges of a lesion and cut out visual noise such as hair occlusion. Malignant lesions usually present with irregular color distribution (brown, red, black, or blue). The variance of Hue (the actual color) and Saturation (the intensity of that color) aid the algorithm in recognizing the atypical

and chaotic color patterns that suggest malignancy [4].

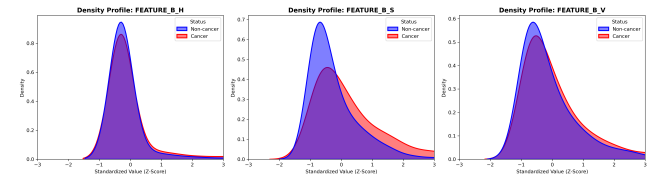


Figure 4: Density profiles of HSV color features across diagnostic classes

Figure 4 shows the kernel density estimates of the three HSV color features (Hue, Saturation, and Value) for cancerous and non-cancerous lesions respectively. All values have been standardized to Z-scores to allow a direct comparison across all features.

The Hue distribution, plot(FEATURE_B_H), depicts a nearly complete overlap between both classes. Both distributions are centered at nearly the same location with similar spread. This informs us that the dominant color tone of a lesion, as a lone statistic, provides limited power in distinguishing malignant from benign lesions. This is to be expected as cancerous and non-cancerous lesions can present with a wide range of color tones. The color of the lesion can depend on the type of lesion, the skin tone of the patient, and the conditions the image was taken in.

The Saturation and Value plots, (FEATURE_B_S and FEATURE_B_V) provide much more information. In both plots, the cancerous distributions show a flatter and heavier right tailed distribution in comparison to the non-cancerous class, especially in the Saturation plot, (FEATURE_B_S). The non-cancerous class is more sharply concentrated at the lower values in both plots. The observed asymmetry in these distributions suggests that malignant lesions in this dataset tend to present with greater color intensity and brightness variation. This is consistent with the irregular pigmentation patterns that are clinically associated with malignancy. As the separation is more apparent in the Saturation plot than in Value, we can discern that the color intensity has a stronger discriminative signal than brightness in this dataset.

Across all three features, there still remains a great deal of overlap between the two classes. This fact reinforces the main argument that color features alone are not sufficient for a reliable classification and the addition of other predictors and feature extraction is necessary to achieve a meaningful separation between cancerous and non-cancerous lesions.

4.2 Hair and Pen marks

When working with real clinical images, we quickly noticed that not everything in the image actually belongs to the lesion. Two things stood out in particular: hair and pen marks. Hair can cover parts of a lesion, which means the colour, border, and shape features we extract may end up describing hair pixels rather than the lesion itself. Pen marks are lines drawn by clinicians around the lesion before taking the photo, and while they are useful in a clinical setting, they add artificial edges and blue colour signals that can confuse our feature extraction. Since both of these appear consistently across many images in the dataset, we decided to address them as

part of our preprocessing pipeline.

4.2.1 Pen removal

Pen marks appear in a subset of images as blue or dark blue markings drawn around the lesion boundary. These marks introduce artificial colour signals and irregular edges that can distort both the border and colour features extracted from the lesion region.

To remove them, we filtered pixels whose colour fell within a defined blue range in the HSV colour space. Blue pen marks occupy a relatively narrow and distinct Hue range that does not commonly appear in natural skin or lesion tissue, making colour-based filtering a reliable approach. Pixels identified as pen marks were masked and removed from the image prior to feature extraction, ensuring that the extracted colour and border features reflected the lesion itself rather than the clinical annotation.



Figure 5: Comparison of the original image and the image after pen removal.

4.2.2 Hair feature

For this feature, we estimated the amount of hair present in each grayscale image by calculating a hair coverage value. A small cross-shaped kernel of 5×5 was created. This was used to apply both top-hat and black-hat morphological transformations. The top-hat operation highlights small bright structures, while the black-hat operation highlights small dark structures, such as dark hairs on lighter skin. The sums of the top-hat and black-hat images were compared to determine which transformation contained the stronger response. However, the final hair mask was created using the black-hat image, since dark hairs were the main structures we wanted to detect. A fixed threshold of 10 was then applied to convert the detected hair regions into a binary mask. In this mask, white pixels represented detected hair and black pixels represented the background. The number of white pixels was counted and divided by the total number of pixels in the image. This produced a hair coverage ratio, representing the proportion of the image covered by detected hair. The value was rounded to four decimal places and used as the hair feature.

4.2.3 Hair removal

Hair removal was performed automatically using morphological filtering and image inpainting. First, the hair coverage value was calculated using the previously described hair feature function. This value was then used to decide whether hair removal was necessary and which kernel size should be

applied. If the hair coverage was below 0.005, the amount of detected hair was considered insignificant, so hair removal was skipped to avoid unnecessary changes to the image.

- **Coverage below 0.005:** Hair coverage was considered insignificant, so hair removal was skipped to avoid unnecessary changes to the image.
- **Coverage between 0.005 and 0.035:** A 12×12 kernel was used, which was suitable for images with a moderate amount of hair.
- **Coverage above 0.035:** A larger 25×25 kernel was used to better detect and remove thicker or more visible hair structures.

After choosing the kernel size, top-hat and black-hat morphological transformations were applied to the grayscale image. These transformations helped highlight thin bright and dark structures in the image. The transformation with the stronger response was selected by comparing the sums of the top-hat and black-hat images. A fixed threshold of 10 was then applied to create a binary hair mask. This mask was used with the original colour image in OpenCV's Telea inpainting method, which replaced the detected hair pixels using information from the surrounding skin and lesion area. This allowed the detected hairs to be removed while preserving as much of the original lesion structure as possible.

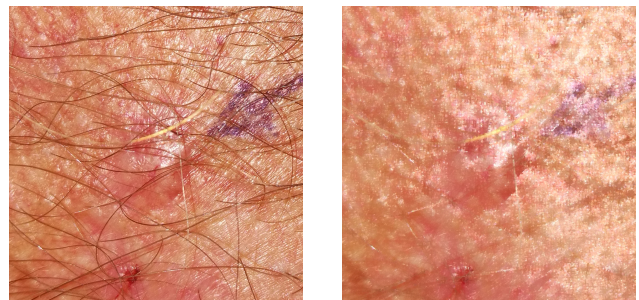


Figure 6: Comparison of the original image and the image after hair removal.

4.3 Best baseline model

In order to find the model that fits our dataset best, we tested our dataset with 2 different models, using cross-validation to find the most suitable parameters for our models:

- KNN model with different k values, in order to find the best k value.
- Random Forest model with different n -estimators and different max-depth values.

For both of our models, first we divided our dataset into two parts: a development set that is 80 percent of the data and a test set that is 20 percent of the dataset. We made sure that the development set and our test set have the random seed of 0. We also used a 5-fold cross-validation strategy to find out which parameters fit our models the best.

4.3.1 K-Nearest Neighbours

KNN allocates a new data point based on its k closest neighbors, then assigning it to the majority class, which in our case,

Cancerous. Then it calculates the probability that the sample belongs to the Malignant group by taking the percentage of the pixels in k range that are also in 'cancerous' group.

We selected the k that gave us the highest mean($k=50$).Figure 7 shows the Cross validation results of our KNN model.

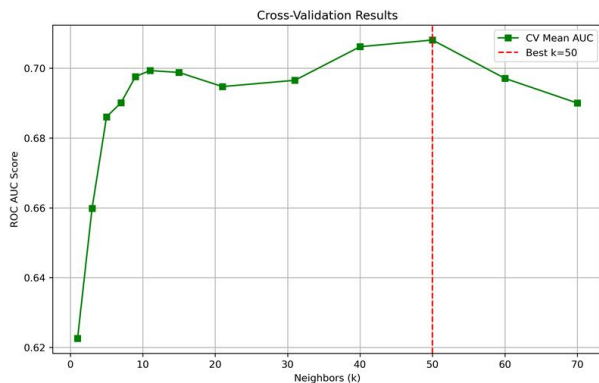


Figure 7: KNN Cross Validation Graph

Then we used this number to train the model again on all development data. on our training data, which gave us a ROC AUC score of 0.7057.

4.3.2 Random Forest

Random Forest method is a way to make decision trees better by making small trees and combining them. This method helps because it stops one tree from being too good and affects the other trees with less impact.

When we use Random Forest to make a prediction for something like whether a lesion is Cancerous or not, it looks at what each tree thinks and averages them out. We tried different ways to make the Random Forest method work best by tuning its hyperparameters using cross validation. First, we looked at the maximum depth of individual trees.

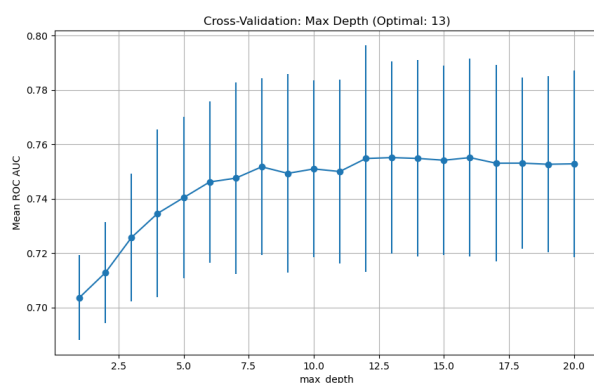


Figure 8: CV Maximum Depth plot for Random Forest Model.

The cross-validation plot for maximum depth indicates that the performance of the model increases gradually as the trees become deeper, finally stabilizing at an optimal depth of 13. If trees are shallower than this, they cannot capture the complexity of the data, and allowing them to grow deeper does not improve performance significantly.

We also experimented with how many trees should be in the forest. As can be observed in the cross-validation estimators

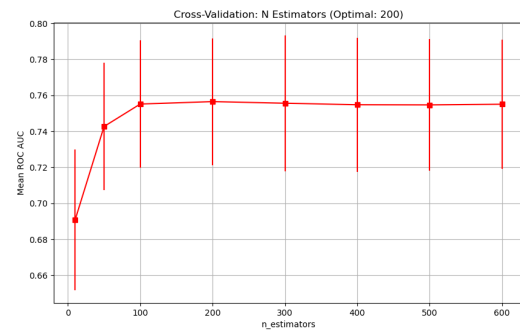


Figure 9: CV n Estimators Plot for Random Forest Model

plot, the model's accuracy increases quickly as more trees are added, but soon plateaus at an optimal number of 200 estimators. Adding more trees beyond 200 simply increases the run time without improving the predictions. Having found these best settings, we built the model and asked it to keep track of its performance. Finally the model achieved the scores below:

- Accuracy: 0.6857
- Recall: 0.7524
- AUC: 0.7370
- Precision: 0.6639
- F1 Score: 0.7054

ROC score of 0.7370 which is shown in Figure 10. We also produced a Confusion Matrix as shown in Figure 11.

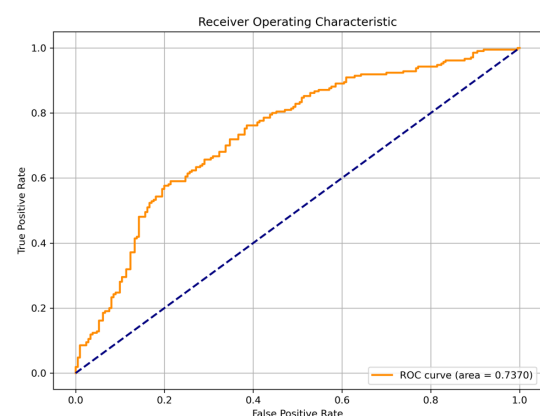


Figure 10: ROC Curve for the Random Forest model.

Confusion Matrix is about Predicted and true labels. It correctly said 158 tumors were malignant and 130 were benign. It made some mistakes such as it said 80 tumors were malignant, when they were not. It said 52 tumors were benign when they were actually malignant.

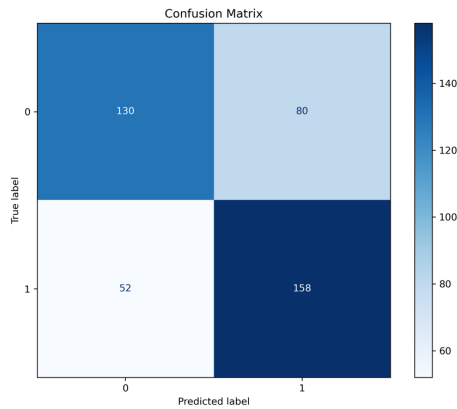


Figure 11: Confusion Matrix for the Random Forest model.

4.3.3 Model Training and Evaluation

Our final predictive model was selected by comparing the final test performances of our two optimised architectures, K-Nearest Neighbours (kNN) and a Random Forest ensemble. Both models were able to differentiate malignant and benign lesions, however, the Random Forest model had better overall performance and better diagnostic balance. The kNN method, which classifies data points according to spatial proximity, was less reliable for this dataset and had a much higher rate of false positives compared to the ensemble method. The Random Forest on the other hand provided much more robust predictive capability. This ensemble method combines several constrained decision trees to decrease the risk of overfitting and to learn better the complex non-linear patterns that discriminate skin lesions. In a clinical setting, it is highly undesirable to generate any false negatives (predict a malignant lesion as benign), but the Random Forest architecture ultimately achieved a more favourable balance between sensitivity and specificity.

We chose Random Forest as our final classification model since it had much stronger generalisation on unseen data and minimised false predictions better than kNN.

5 Open Question: Evaluating the Role of LBP Texture Features and SHAP Explanations in Skin Lesion Classification

After evaluating the baseline model, we found that some of the traditional ABC features, particularly asymmetry and border compactness, did not clearly separate cancerous from non-cancerous lesions. The feature distributions overlapped considerably between the two classes, suggesting that shape-based features alone may not contain enough diagnostic information. In addition, visual inspection of false negative cases showed that the model often struggled with lesions that had surface abnormalities, such as scaling, roughness, or crust-like texture. This motivated us to investigate whether texture-based features could provide additional information for classification.

To explore this, we used Local Binary Patterns (LBP) as an additional texture descriptor [2]. LBP describes local texture by comparing each pixel with its neighbouring pixels and encoding the result as a binary pattern. In our implementation, each pixel was compared with eight neighbouring pixels. If a neighbouring pixel had an intensity greater than or equal to the central pixel, it contributed to the LBP code. These codes were then collected into a histogram with 256 bins. As a result, LBP added 256 texture-related features to the dataset, representing how frequently each local texture pattern appeared in the lesion image.

LBP was included because it captures small local structures within the lesion. While border features describe the outer shape of the lesion and colour features describe pigmentation, LBP focuses on internal texture. This is relevant because malignant lesions may contain irregular surface patterns that are not fully captured by asymmetry, compactness, convexity, or colour statistics. However, adding 256 LBP features also introduced a challenge: high dimensionality. The number of LBP features was much larger than the original ABC feature set, which can make the model more complex and increase the risk of overfitting.

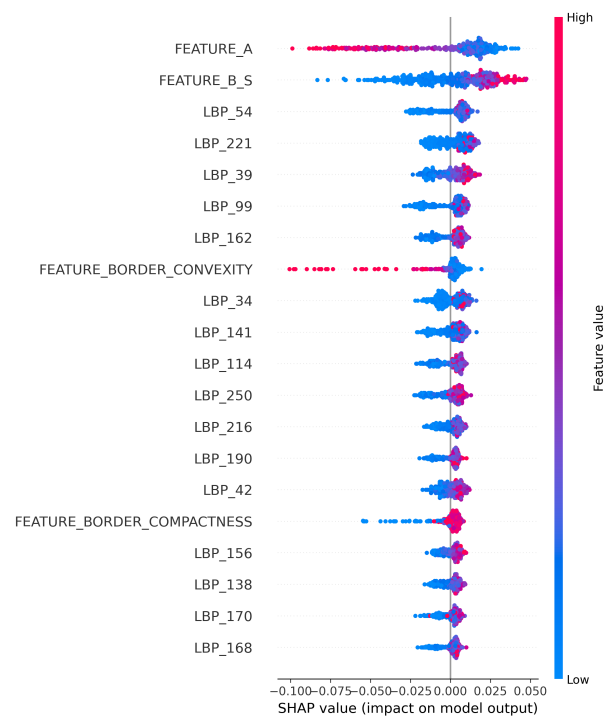


Figure 12: SHAP summary plot for the Random Forest model using ABC and LBP features.

To understand how the Random Forest model used the different features, we applied SHAP analysis. Figure 12 shows the global SHAP summary plot, which illustrates both the importance and direction of each feature across the test set. The plot shows that **FEATURE_A** had the strongest overall influence on the model predictions, followed by **FEATURE_B_S**. Several LBP texture features, including **LBP_54**, **LBP_221**, **LBP_39**, **LBP_99**, and **LBP_162**, also appeared among the highest-ranked features. This suggests that the model did not rely only on the traditional ABC features, but also used texture information extracted through LBP.

In the SHAP summary plot, colour represents the feature value, where blue indicates low values and pink indicates high values. The horizontal position shows whether a feature pushed the prediction toward or away from the cancerous class. **FEATURE_A** had the widest SHAP value spread, meaning that it had the strongest influence on the model's predictions. In this feature, higher values generally pushed the prediction downward, while lower values tended to push the prediction upward. In contrast, higher values of **FEATURE_B_S** mostly pushed the prediction upward, while lower values tended to reduce the predicted probability. The LBP features generally had smaller individual effects, but their presence among the most important features suggests that texture contributed meaningfully to the classification process.

Border-related features, such as **FEATURE_BORDER_CONVEXITY** and **FEATURE_BORDER_COMPACTNESS**, also appeared in the SHAP plot. However, they were ranked below the strongest asymmetry-related and texture-based features. This is important because the earlier analysis showed that the distributions of border and asymmetry features overlapped between cancerous and non-cancerous lesions. The SHAP results therefore suggest that, while these traditional features were still used by the model, the LBP texture features provided additional information that helped explain some predictions.

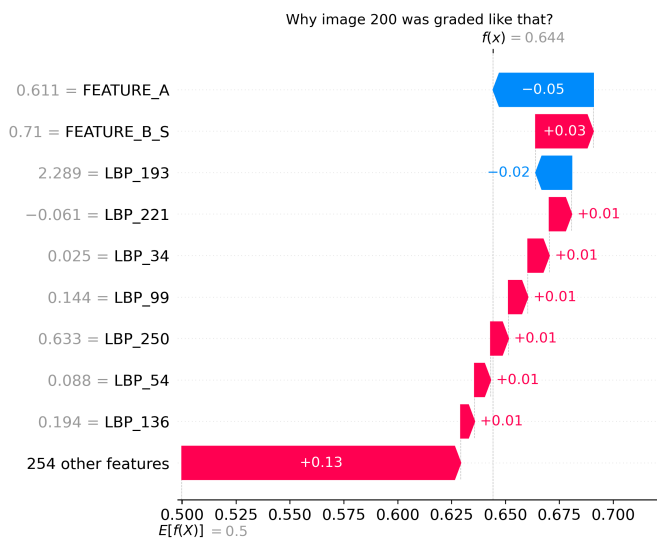


Figure 13: SHAP waterfall plot explaining the prediction for image 200.

To examine an individual prediction, we generated a SHAP waterfall plot for image 200, shown in Figure 13. The expected model output was approximately $E[f(X)] = 0.5$, while the final prediction for this image was $f(x) = 0.644$. This means that the model increased the prediction above the baseline probability and moved it toward the cancerous class.

The waterfall plot shows how each feature contributed to this specific prediction. The largest negative contribution came from **FEATURE_A**, with a value of 0.611, which reduced the prediction by approximately -0.05 . This means that, for this image, **FEATURE_A** made the model less likely to predict cancer. Another negative contribution came from **LBP_193**, which reduced the prediction by approximately

-0.02 . However, **FEATURE_B_S**, with a value of 0.71, increased the prediction by approximately $+0.03$. Several LBP features, including **LBP_221**, **LBP_34**, **LBP_99**, **LBP_250**, **LBP_54**, and **LBP_136**, also made small positive contributions of around $+0.01$ each. Although each individual LBP contribution was relatively small, their combined effect helped push the final prediction upward. The group labelled “254 other features” also made a notable positive contribution of approximately $+0.13$, showing that many smaller features collectively influenced the final output.

This individual explanation is useful because it shows that the model's decision was not based on a single feature. Instead, the final prediction resulted from a combination of traditional ABC-related features and several LBP texture features. In this case, **FEATURE_A** reduced the predicted probability, but this was outweighed by the positive contribution of **FEATURE_B_S**, several LBP features, and the combined effect of the remaining features. Overall, the SHAP results suggest that LBP features can provide useful additional information, but their contribution is spread across many individual bins rather than concentrated in one easily interpretable feature. This makes LBP valuable for machine learning, but more difficult to explain clinically. For example, it is difficult to visually interpret what a feature such as **LBP_54** or **LBP_221** represents without first mapping those codes back to specific texture patterns in the image.

Overall, LBP allowed us to investigate lesion texture beyond the traditional ABC features, while SHAP helped explain how both traditional and texture-based features influenced the model's predictions.

6 Results

The results show that the extracted ABC-based features were able to provide some separation between cancerous and non-cancerous lesions, but the classes still showed considerable overlap. In particular, the asymmetry and border features did not clearly distinguish the two groups when examined individually, suggesting that shape-based features alone were not sufficient for reliable classification. The Random Forest model performed better than the K-Nearest Neighbours model overall, as it was able to capture more complex relationships between the extracted features. The confusion matrix showed that the model correctly classified many cancerous and non-cancerous lesions, but it still produced both false positives and false negatives. The false negatives are especially important in this task, because they represent cancerous lesions that were predicted as non-cancerous. This indicates that while the model can support skin lesion classification, it would not be reliable enough for clinical use on its own. The addition of LBP texture features and SHAP analysis provided further insight into the model's decisions, showing that texture information contributed alongside the traditional ABC features. Overall, the results suggest that interpretable features such as asymmetry, border irregularity, colour variation, and texture can form a useful foundation for classification, but further improvements in segmentation quality, feature extraction, and

model validation are needed to improve reliability.

6.1 Limitations and Improvements

Although the pipeline developed in this project demonstrates that ABC features can support binary skin lesion classification, several limitations affect the reliability of the results. The features we extracted from the lesion images depend entirely on the accuracy of the segmentation masks. The Asymmetry scores and the border measurements (compactness and convexity) are directly computed from the shape of the mask. If imprecise or noisy masks are present, our algorithm will produce distorted feature values regardless of what the lesion truly looks like. The only quality control that was applied to the masks was to exclude completely blank/black masks. Masks that were partially incorrect or poorly aligned with the lesion boundary have likely introduced noise into the data without being detected.

We intentionally selected features to be easily interpretable, but these features lack descriptive power. Asymmetry, border compactness, border convexity, and HSV variance statistics only capture a subset of visual information that is fully present in each lesion image. Our SHAP analysis showed that the LBP texture features add meaningful contributions to the Random Forest predictions. This suggests that the surface patterns such as, scaling, roughness, and crusting contain diagnostic information that the ABC features can not fully obtain.

The addition of LBP features caused a significant increase in the dimensionality of the feature space. We originally started with a small set of hand-crafted ABC features, and when the LBP was introduced we added 256 additional features per image. We have a risk of overfitting from this addition as we create disproportion between the number of features and number of training samples. Our model then has the possibility of capturing correlations present in the training data that do not generalize to unseen examples. Our Random Forest Model is inherently more robust towards higher dimensionality than simpler classifiers because of the random subspace sampling applied at each split. We can reduce the risk of overfitting but not fully eliminate it.

A more rigorous and ultimately better approach would involve applying Principal Component Analysis (PCA), to reduce the dimensionality. PCA projects the 256 LBP bins onto a lower-dimensional subspace. This method helps to retain the majority of the variance while simultaneously discarding the noise. In the absence of this technique, we cannot fully rule out that some of our model's predictive power reflects overfitting to the LBP noise. Also, while LBP features have improved model performance, it becomes difficult to explain them in a clinical context. It would be simple for a dermatologist to explain to a clinician, the impact of asymmetry or irregular borders, but a feature such as LBP-114 has no visual equivalent to be communicated. Another potential improvement would be to replace or supplement the current masks with automatically generated segmentation. We could use a dedicated model to do this which would reduce the effect of the mask noise on feature quality.

Also something to note, between the cancerous and non-

cancerous images, the split was nearly equal with 1048 cancerous and 1055 non-cancerous images (Table 2). The distribution within the cancerous group was heavily skewed, where melanoma only accounted for 49 of the images compared to the 815 for basal cell carcinoma from the 2103 images in the dataset (Table 1). Our model was trained on the binary label, cancerous vs. non-cancerous, so it had limited opportunity to learn the specific visual patterns of melanoma. Clinically speaking, this is a major drawback since melanoma is known to be the most dangerous form of skin cancer due to high metastatic potential. Early detection is crucial for melanoma patient outcomes. Our model which has been trained on fewer than 50 melanoma lesion examples in training will correctly classify the BCC lesions that were predominant in the data, but will systematically fail to detect melanoma. This affect will not have a major effect on the total AUC score, but would have severe consequences in a clinical deployment of our model. To improve this, it would be necessary to incorporate many more melanoma images to help train the model more effectively.

7 Conclusion

For this project, we developed a machine learning pipeline for classifying skin lesions as cancerous or non-cancerous using interpretable image features extracted from the PAD-UFES-20 dataset. The dataset was first explored to understand class balance, diagnostic distribution, missing values, and mask availability. The diagnostic labels were then converted into a binary target, producing an almost balanced classification task.

The Random Forest model outperformed KNN across all metrics, achieving an AUC of 0.7370, accuracy of 0.6857, and recall of 0.7524. The results show that hand crafted features can provide a reasonable separation between the two classes, though the model still misclassified 52 cancerous lesions as benign, which is the type of error we want to avoid in this context

Feature analysis and SHAP results show that asymmetry (Feature A) and color saturation (Feature B) were the strongest predictors, while border features contributed but ranked lower in the metric. Adding LBP texture features helped the model pick up on surface patterns like scaling and roughness that the ABC features alone missed.

Removing hair and pen marks from images before feature extraction is an important preprocessing step, since both can interfere with the colour and border features we extract from the lesion. However, based on our results we were not able to clearly measure how much this actually improved classification performance.

Overall, the project shows that ABC-inspired features provide a useful foundation for skin lesion classification. However, the results depend strongly on segmentation quality, feature design, and the limited number of samples for some diagnoses. Future work should focus on improving mask quality, adding stronger texture and colour descriptors, and testing whether

metadata or deep learning methods can improve performance.

8 Appendix: Additional Figures

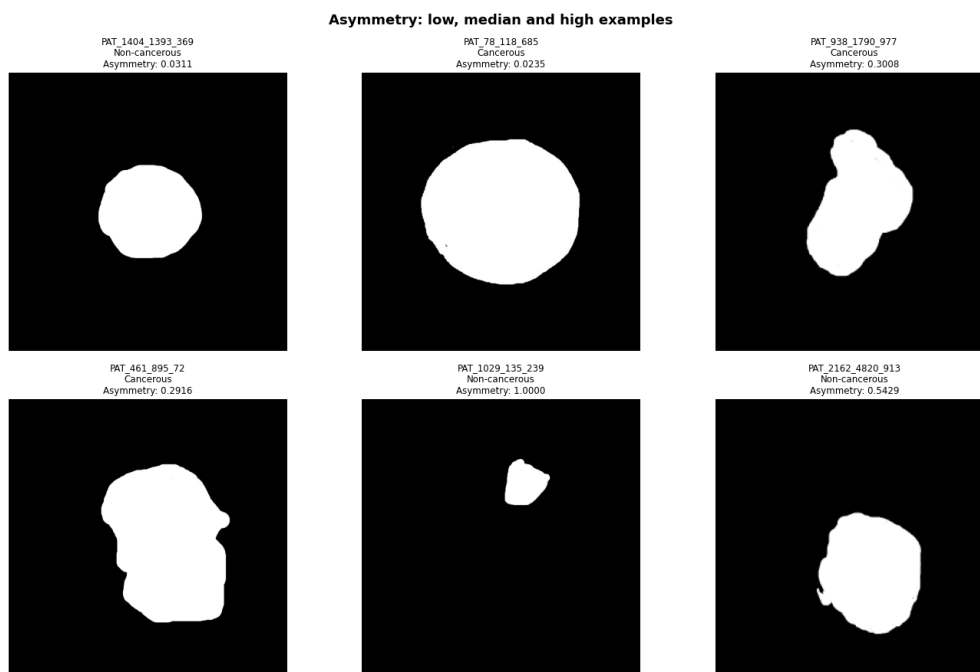


Figure 14: Asymmetry scores for cancerous and non-cancerous lesions.

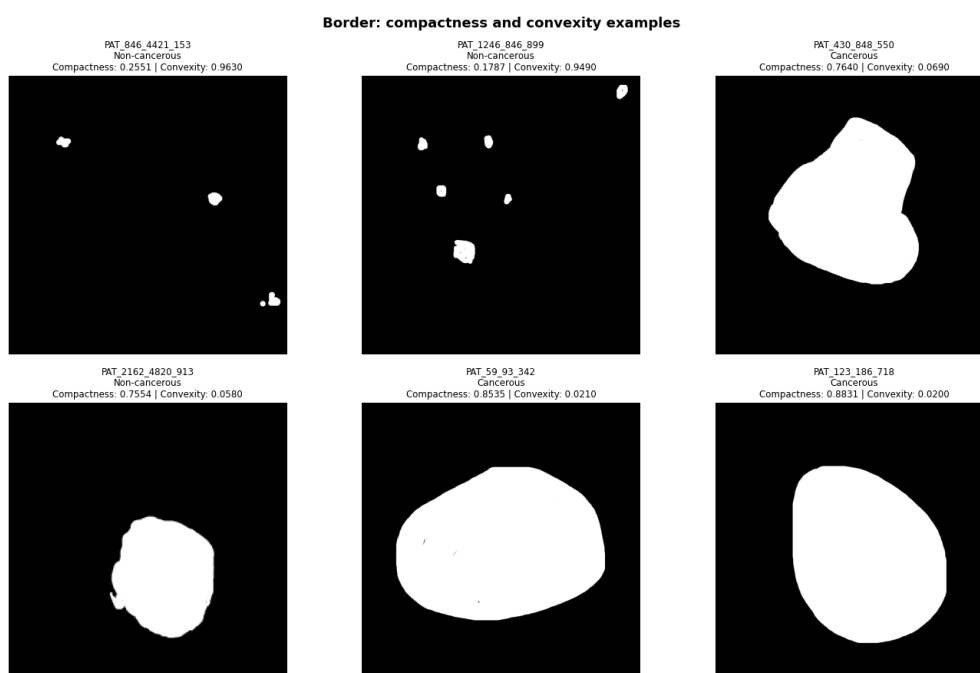


Figure 15: Border feature scores for cancerous and non-cancerous lesions.

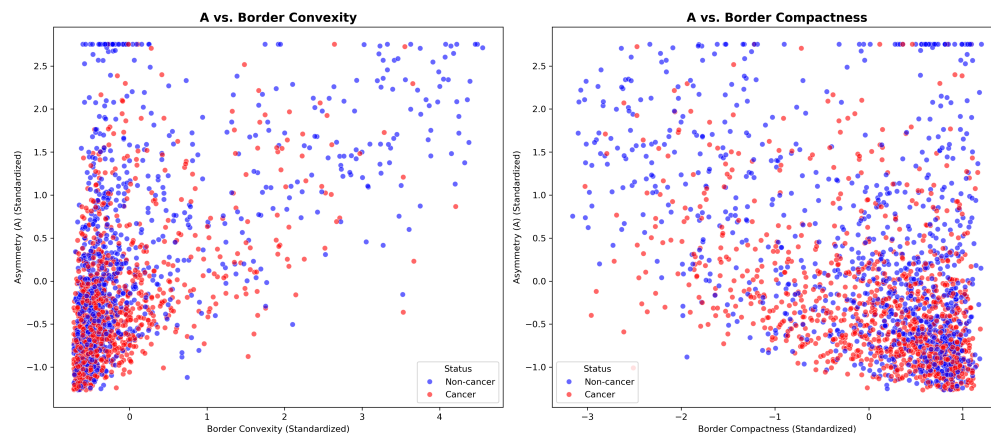


Figure 16: Relationship between asymmetry and border features.

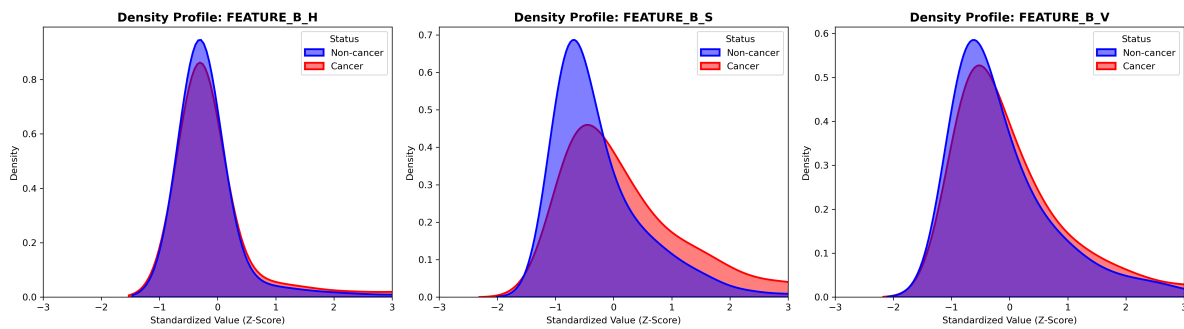


Figure 17: Density profiles of HSV colour features across diagnostic classes.



Figure 18: Comparison of an original image and the result after pen mark removal.

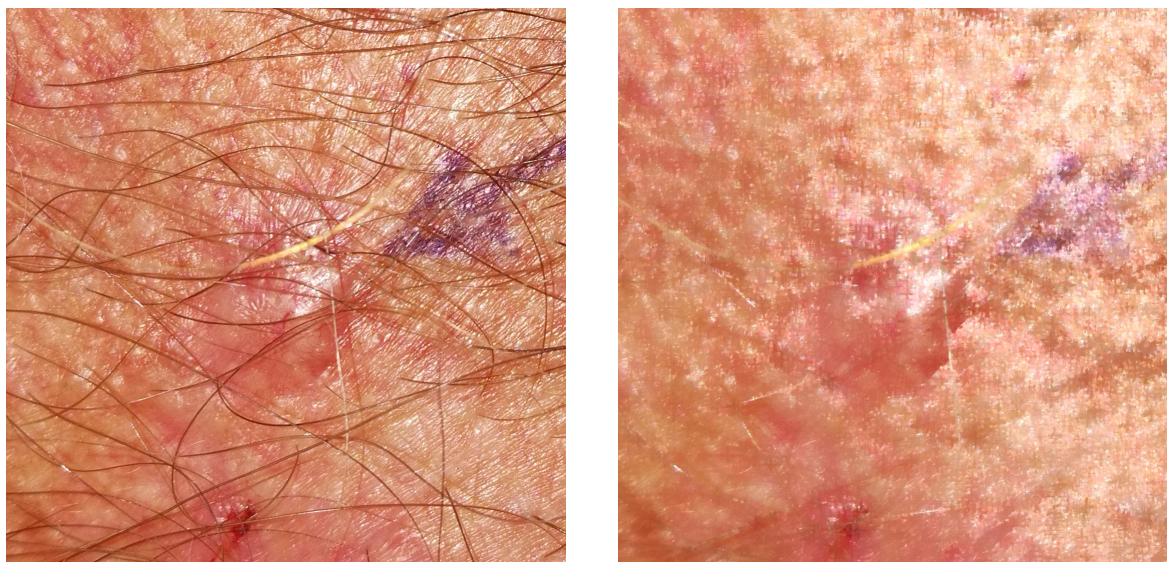


Figure 19: Comparison of an original image and the result after hair removal.

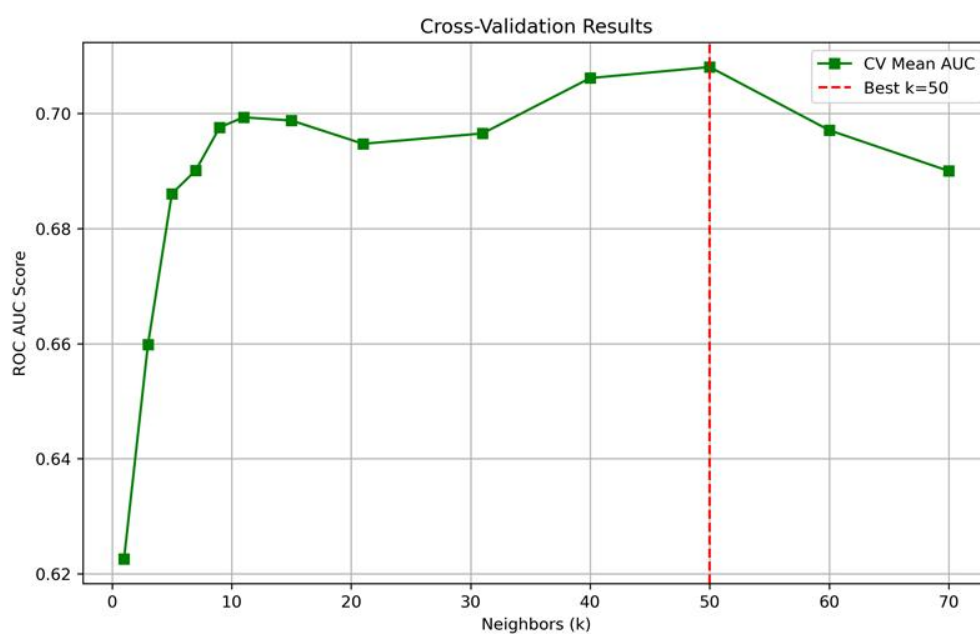


Figure 20: Cross-validation results for the KNN model.

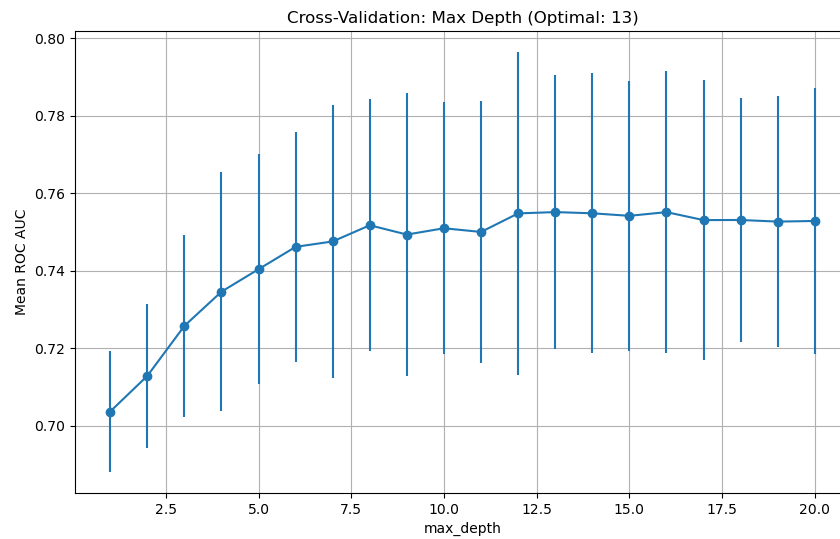


Figure 21: Cross-validation results for maximum depth in the Random Forest model.

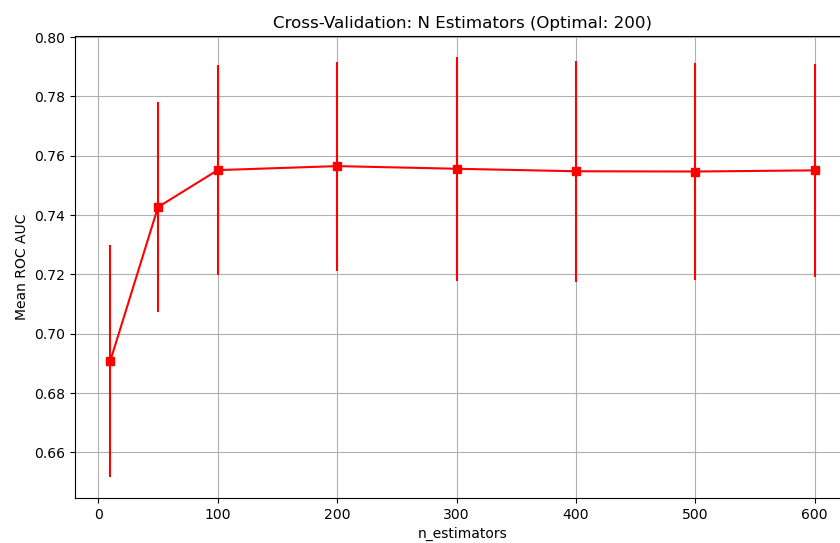


Figure 22: Cross-validation results for the number of estimators in the Random Forest model.

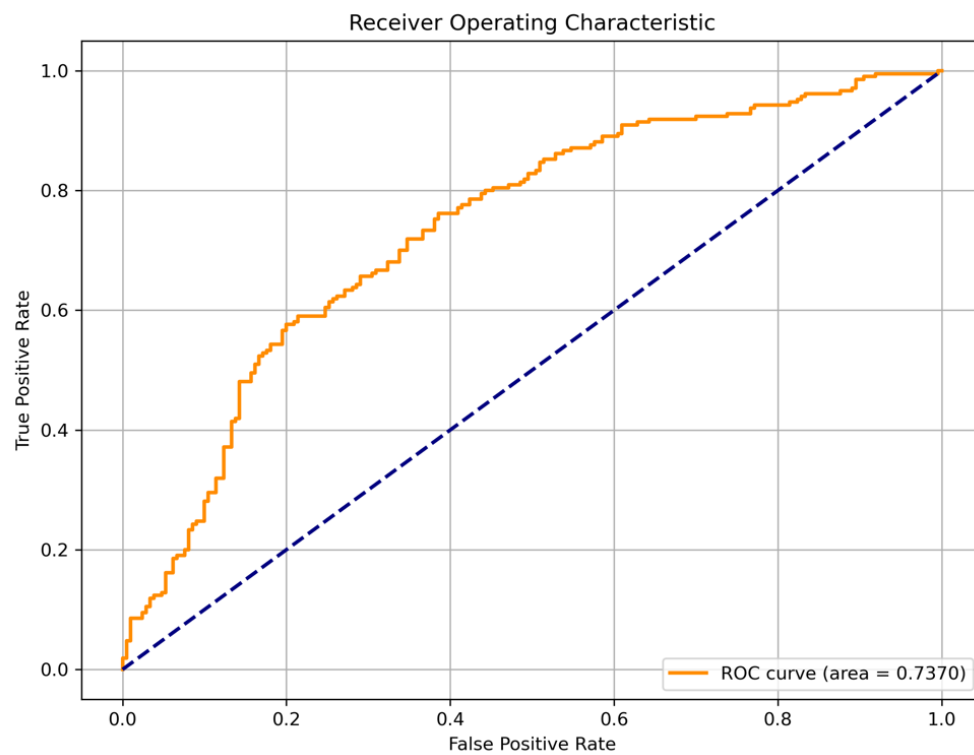


Figure 23: ROC curve for the Random Forest model.

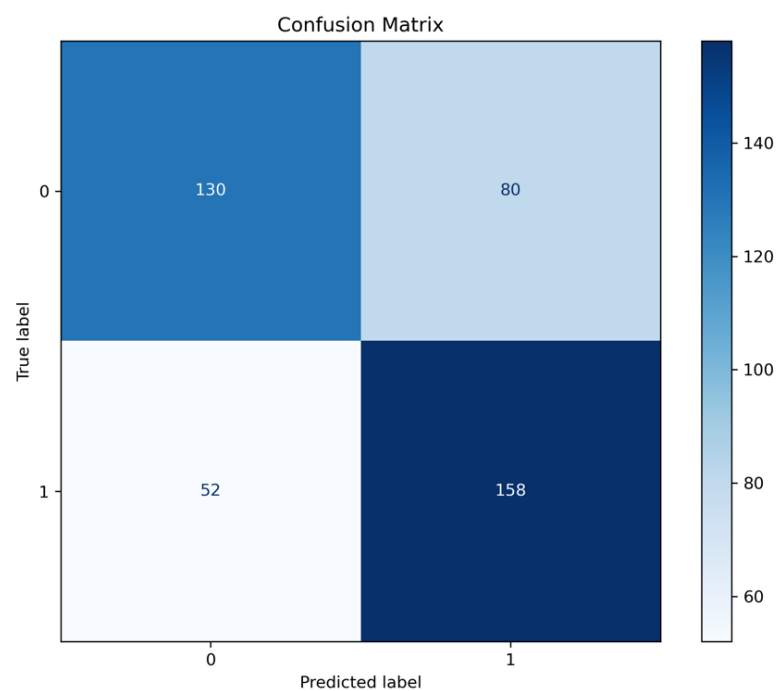


Figure 24: Confusion matrix for the Random Forest model.

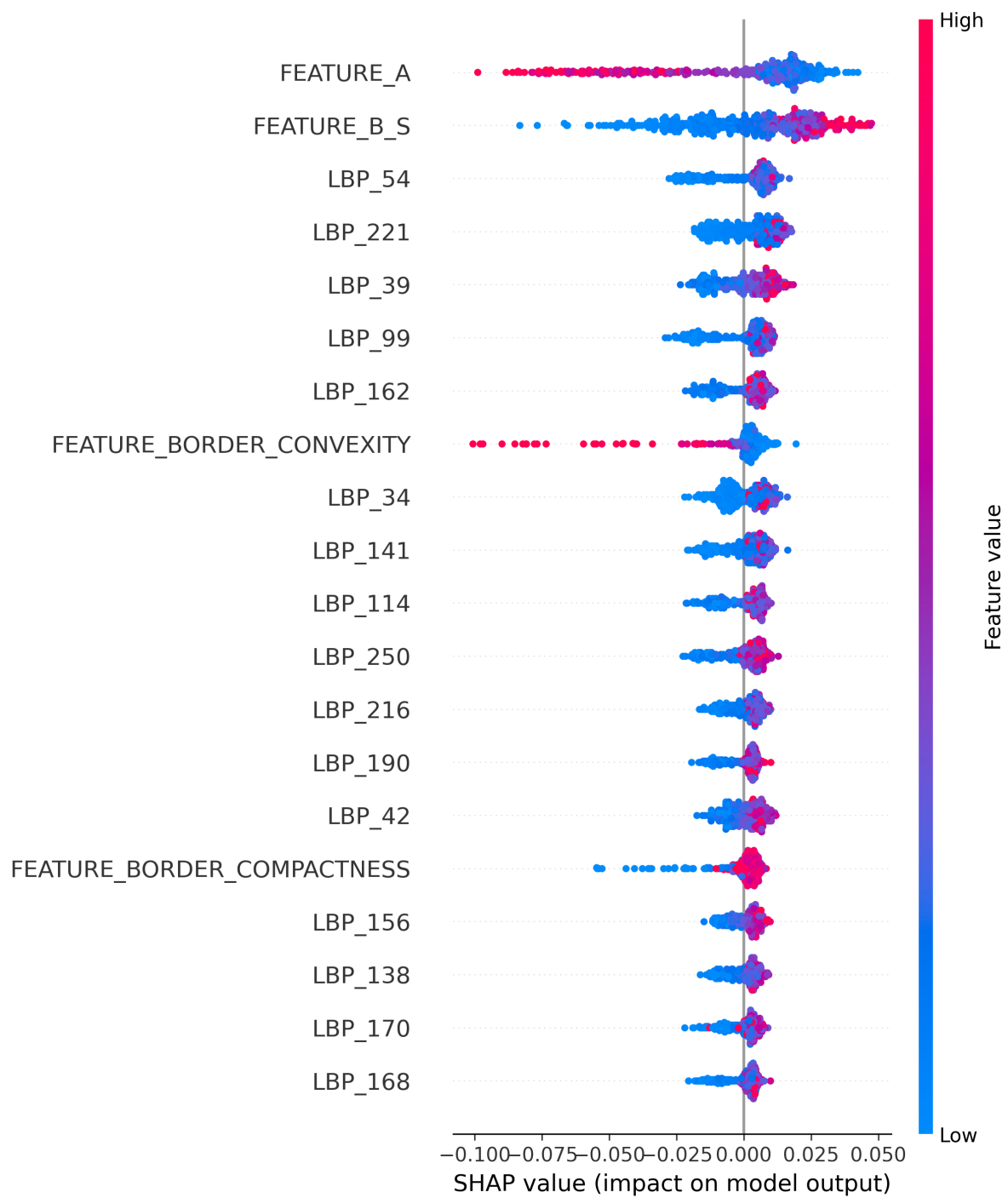


Figure 25: SHAP summary plot for the Random Forest model using ABC and LBP features.

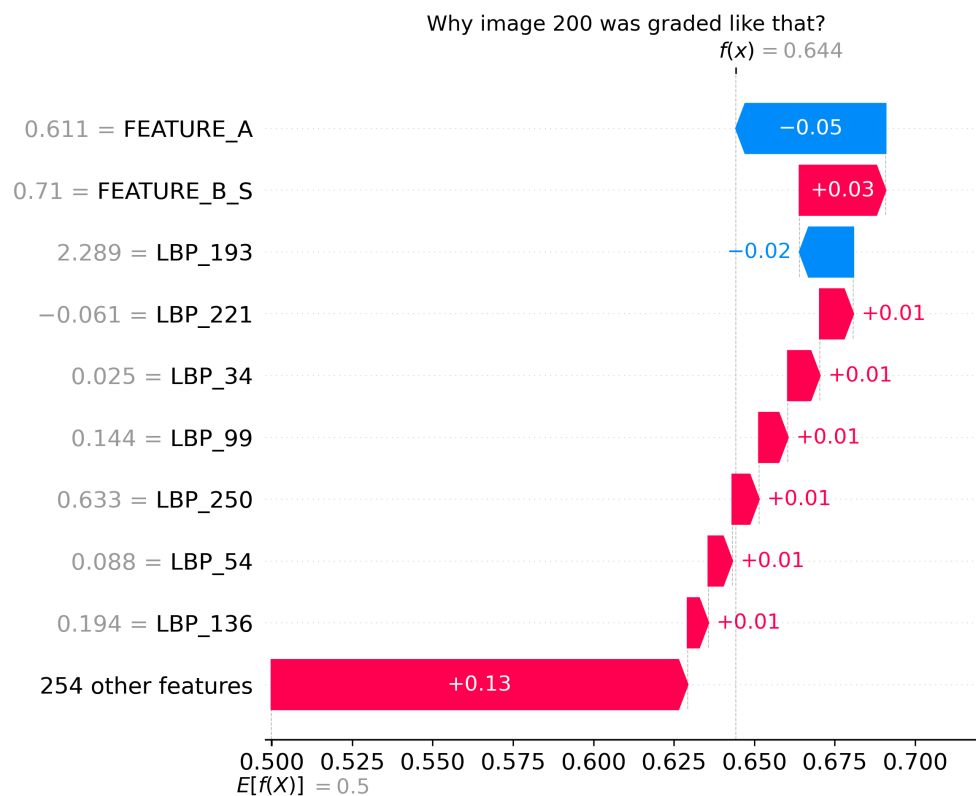


Figure 26: SHAP waterfall plot explaining the prediction for image 200.

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